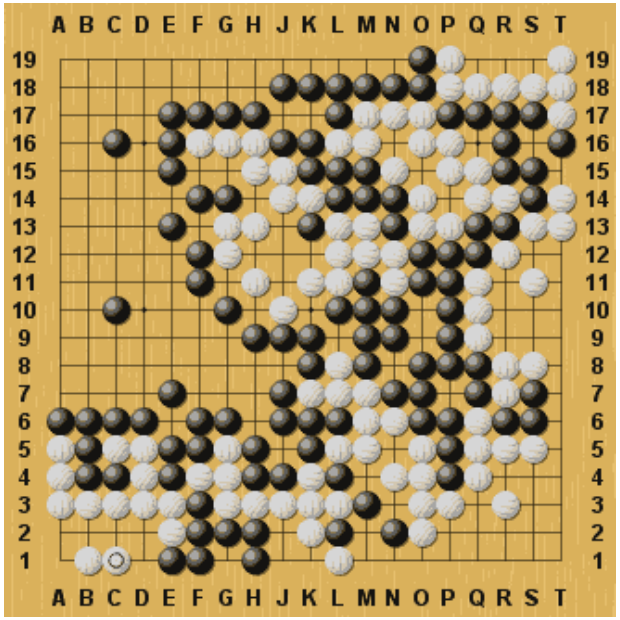
ML:Reinforcement learning

Game Playing

- Example: Go (one of the oldest and hardest board games)



- **Agent:**
- **Environment:**
- **State:**
- **Action:**
- **Reward**

- 2016: AlphaGo defeats top player Lee Sedol (4-1)
  - Game 2 move 37: AlphaGo plays unexpected move (odds 1/10,000)

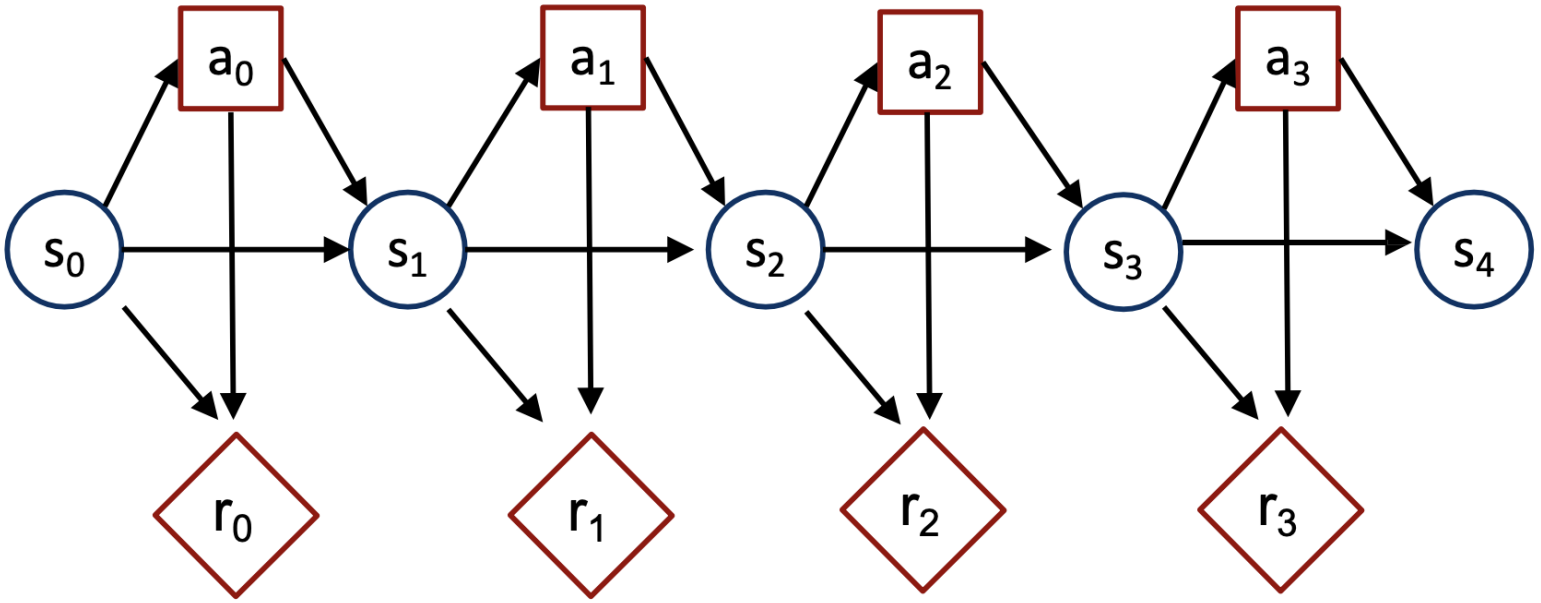
Markov Decision Process

- Definition
  - Set of states: S
  - Set of actions: A
  - Transition model: $\Pr(s_t|s_{t-1}, a_{t-1})$
  - Reward model: $R(s_t, a_t)$
  - Discount factor: $0 \leq \gamma \leq 1$
    - discounted: $\gamma < 1$ undiscounted: $\gamma = 1$
  - Horizon (i.e., # of time steps): h
    - Finite horizon: $h \in \mathbb{N}$ infinite horizon: $h = \infty$

- Goal: find optimal policy

Markov Decision Process

- Markov process augmented with…
  - Actions e.g., a_t
  - Rewards e.g., r_t



Current Assumptions

- Uncertainty: stochastic process
- Time: sequential process
- Observability: fully observable states (observe the environment completely and exactly)
- No learning: complete model
- Variable type: discrete (e.g., discrete states and actions)

Rewards

- **Rewards**: $r_t \in \Re$
- **Reward function**: $R(s_t, a_t) = r_t$
  mapping from state-action pairs to rewards
- Common assumption: stationary reward function
  - $R(s_t, a_t)$ is the same $\forall t$ (function)
- Exception: terminal reward function often different
  - E.g., in a game: 0 reward at each turn and +1/-1 at the end for winning/losing
- Goal: **maximize sum of rewards** $\sum_t R(s_t, a_t)$

Discounted/Average Rewards

- If process infinite, isn't $\sum_t R(s_t, a_t)$ infinite?

- Solution 1: discounted rewards
  - Discount factor: $0 \leq \gamma < 1$
  - Finite utility: $\sum_t \gamma^t R(s_t, a_t)$ is a geometric sum
  - γ induces an inflation rate of $1/\gamma - 1$
  - Intuition: prefer utility sooner than later

- Solution 2: average rewards
  - More complicated computationally
  - Beyond the scope of this course

Markov Decision Process

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Policy

- Choice of action at each time step

- Formally:
  - Mapping from states to actions
  - i.e., $\pi(s_t) = a_t$
  - Assumption: fully observable states
    - Allows a_t to be chosen only based on current state s_t

Policy Optimization

- Policy evaluation:
  - Compute expected utility

$$V^{\pi}(s_0) = \sum_{t=0}^{h} \gamma^t \sum_{s_t} \Pr(s_t | s_0, \pi)\, R(s_t, \pi(s_t))$$

- Optimal policy:
  - Policy with highest expected utility

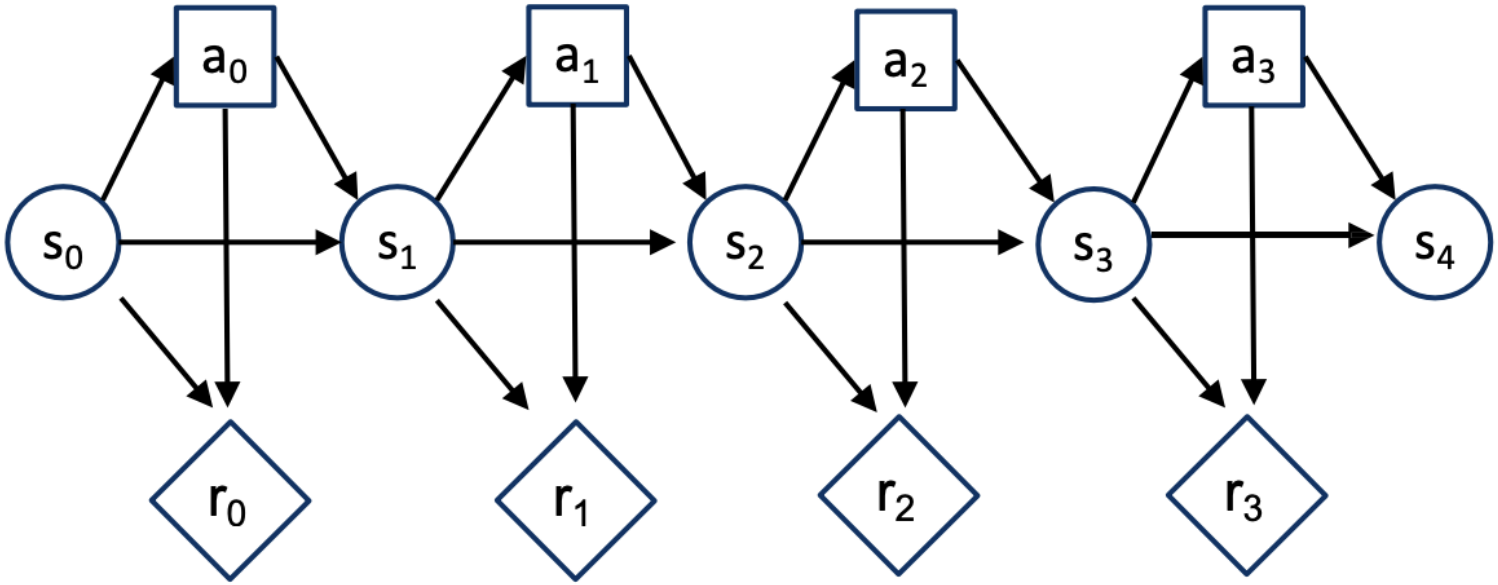
$$V^{\pi^*}(s_0) \geq V^{\pi}(s_0) \quad \forall \pi$$

Policy Optimization

- Several classes of algorithms:
  - Value iteration
  - Policy iteration
  - Linear Programming
  - Search techniques
- Computation may be done
  - Offline: before the process starts
  - Online: as the process evolves

Value Iteration

- Performs dynamic programming
- Optimizes decisions in reverse order



Value Iteration

- Value when no time left:

$$V(s_h) = \max_{a_h} R(s_h, a_h)$$

- Value with one time step left:

$$V(s_{h-1}) = \max_{a_{h-1}} R(s_{h-1}, a_{h-1}) + \gamma \sum_{s_h} \Pr(s_h | s_{h-1}, a_{h-1})\, V(s_h)$$

- Value with two time steps left:

$$V(s_{h-2}) = \max_{a_{h-2}} R(s_{h-2}, a_{h-2}) + \gamma \sum_{s_{h-1}} \Pr(s_{h-1} | s_{h-2}, a_{h-2})\, V(s_{h-1})$$

- …

- Bellman's equation:

$$V(s_t) = \max_{a_t} R(s_t, a_t) + \gamma \sum_{s_{t+1}} \Pr(s_{t+1} | s_t, a_t)\, V(s_{t+1})$$

$$a_t^* = \arg\max_{a_t} R(s_t, a_t) + \gamma \sum_{s_{t+1}} \Pr(s_{t+1} | s_t, a_t)\, V(s_{t+1})$$

Value Iteration Algorithm

valueIteration(MDP)

$V_0^*(s) \leftarrow \max_a R(s,a) \ \forall s$

For $t = 1$ to h do

$V_t^*(s) \leftarrow \max_a R(s,a) + \gamma \sum_{s'} \Pr(s'|s,a) V_{t-1}^*(s') \ \forall s$

Return V^*

Optimal policy π^*

$t = 0$: $\pi_0^*(s) \leftarrow \operatorname{argmax}_a R(s,a) \ \forall s$

$t > 0$: $\pi_t^*(s) \leftarrow \operatorname{argmax}_a R(s,a) + \gamma \sum_{s'} \Pr(s'|s,a) V_{t-1}^*(s') \ \forall s$

NB: t indicates the # of time steps to go (till end of process)

π^* is non stationary (i.e., time dependent)

Value Iteration

- Matrix form:
  - R^a: $|S| \times 1$ column vector of rewards for a
  - V_t^*: $|S| \times 1$ column vector of state values
  - T^a: $|S| \times |S|$ matrix of transition prob. for a

valueIteration(MDP)

$V_0^* \leftarrow \max_a R^a$

For $t = 1$ to h do

$V_t^* \leftarrow \max_a R^a + \gamma T^a V_{t-1}^*$

Return V^*

Policy Optimization

- Value iteration
  - Optimize value function
  - Extract induced policy

- Can we directly optimize the policy?
  - Yes, by policy iteration

Policy Iteration

- Alternate between two steps

1. Policy evaluation

$$V^{\pi}(s) = R\big(s, \pi(s)\big) + \gamma \sum_{s'} \Pr(s'|s, \pi(s)) V^{\pi}(s') \qquad \forall s$$

2. Policy improvement

$$\pi(s) \leftarrow \underset{a}{\operatorname{argmax}}\, R(s, a) + \gamma \sum_{s'} \Pr(s'|s, a)\, V^{\pi}(s') \qquad \forall s$$

Algorithm

policyIteration(MDP)

Initialize π_0 to any policy

$n \leftarrow 0$

Repeat

Eval: $V_n = R^{\pi_n} + \gamma T^{\pi_n} V_n$

Improve: $\pi_{n+1} \leftarrow argmax_a R^a + \gamma T^a V_n$

$n \leftarrow n + 1$

Until $\pi_{n+1} = \pi_n$

Return π_n

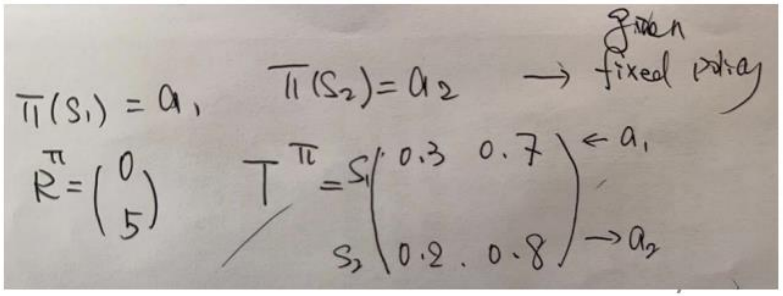
Policy evaluation

- Linear system of equations

$$V_{\infty}^{\pi}(s) = R\big(s, \pi_{\infty}(s)\big) + \gamma \sum_{s'} \Pr(s'|s, \pi_{\infty}(s))\, V_{\infty}^{\pi}(s')\ \forall s$$

- Matrix form:
  - R:$|S|\times 1$ column vector of sate rewards for π
  - V:$|S|\times 1$ column vector of state values for π
  - T:$|S|\times|S|$ matrix of transition prob for π

$$V = R + \gamma T V$$



Reinforcement Learning

- Definition
  - States: $s \in S$
  - Actions: $a \in A$
  - Rewards: $r \in \mathbb{R}$
  - Transition model: $\Pr(s_t|s_{t-1}, a_{t-1})$
  - Reward model: $\Pr(r_t|s_t, a_t)$

  } unknown model

  - Discount factor: $0 \leq \gamma \leq 1$
    - discounted: $\gamma < 1$ undiscounted: $\gamma = 1$
  - Horizon (i.e., # of time steps): h
    - Finite horizon: $h \in \mathbb{N}$ infinite horizon: $h = \infty$

- Goal: find optimal policy π^* such that

$$\pi^* = argmax_\pi \sum_{t=0}^{h} \gamma^t E_\pi[r_t]$$

Policy optimization

- Markov Decision Process:
  - Find optimal policy given transition and reward model
  - Execute policy found
- Reinforcement learning:
  - Learn an optimal policy while interacting with the environment

Important Components in RL

RL agents may or may not include the following components:

- Model: $\Pr(s'|s,a), \Pr(r|s,a)$
  - Environment dynamics and rewards
- Policy: $\pi(s)$
  - Agent action choices
- Value function: $V(s)$
  - Expected total rewards of the agent policy

Categorizing RL agents

Value based
- No policy (implicit)
- Value function

Policy based
- Policy
- No value function

Actor critic
- Policy
- Value function

Model based
- Transition and reward model

Model free
- No transition and no reward model (implicit)

Model free evaluation

- Given a policy π, estimate $V^{\pi}(s)$ without any transition or reward model

- **Monte Carlo** evaluation

$$V^{\pi}(s) = E_{\pi}[\sum_{t} \gamma^{t} r_{t}]$$
$$\approx \frac{1}{n(s)} \sum_{k=1}^{n(s)} [\sum_{t} \gamma^{t} r_{t}^{(k)}] \qquad \text{(sample approximation)}$$

- **Temporal difference (TD)** evaluation

$$V^{\pi}(s) = \mathrm{E}[\mathrm{r}|\mathrm{s}, \pi(s)] + \gamma \sum_{s'} \Pr(s'|s, \pi(s))\, V^{\pi}(s')$$
$$\approx \mathrm{r} + \gamma V^{\pi}(s') \qquad \text{(one sample approximation)}$$

Monte Carlo Evaluation

- Let G_k be a one-trajectory Monte Carlo target

$$G_k = \sum_t \gamma^t r_t^{(k)}$$

- Approximate value function

$$V_n^\pi(s) \approx \frac{1}{n(s)} \sum_{k=1}^{n(s)} G_k$$

$$= \frac{1}{n(s)} \left(G_{n(s)} + \sum_{k=1}^{n(s)-1} G_k\right)$$

$$= \frac{1}{n(s)} \left(G_{n(s)} + (n(s)-1) V_{n-1}^\pi(s)\right)$$

$$= V_{n-1}^\pi(s) + \frac{1}{n(s)} \left(G_{n(s)} - V_{n-1}^\pi(s)\right)$$

- **Incremental update**

$$V_n^\pi(s) \leftarrow V_{n-1}^\pi(s) + \alpha_n \left(G_{n(s)} - V_{n-1}^\pi(s)\right)$$

learning rate $1/n(s)$

Temporal Difference Evaluation

- Approximate value function: $V^{\pi}(s) \approx r + \gamma V^{\pi}(s')$
- **Incremental update**

$$V_n^{\pi}(s) \leftarrow V_{n-1}^{\pi}(s) + \alpha_n(r + \gamma V_{n-1}^{\pi}(s') - V_{n-1}^{\pi}(s))$$

- **Theorem:** If α_n is appropriately decreased with number of times a state is visited then $V_n^{\pi}(s)$ converges to correct value
- **Sufficient conditions** for α_n:

(1) $\sum_n \alpha_n \rightarrow \infty$ (2) $\sum_n (\alpha_n)^2 < \infty$

- Often $\alpha_n \ s \ = 1/n(s)$
  - Where $n(s)$ = # of times s is visited

Temporal Difference (TD) evaluation

TDevaluation(π, V^{π})

Repeat

Execute $\pi(s)$

Observe s' and r

Update counts: $n(s) \leftarrow n(s) + 1$

Learning rate: $\alpha \leftarrow 1/n(s)$

Update value: $V^{\pi}(s) \leftarrow V^{\pi}(s) + \alpha(r + \gamma V^{\pi}(s') - V^{\pi}(s))$

$s \leftarrow s'$

Until convergence of V^{π}

Return V^{π}

Model Free Control

- Instead of evaluating the state value fn, $V^{\pi}(s)$, evaluate the state-action value fn, $Q^{\pi}(s,a)$

$Q^{\pi}(s,a)$: value of executing a followed by π

$$Q^{\pi}(s,a) = E[r|s,a] + \gamma \sum_{s'} \Pr(s'|s,a) V^{\pi}(s')$$

- Greedy policy π'

$$\pi'(s) = argmax_a Q^{\pi}(s,a)$$

Bellman's Equation

- Optimal state value function $V^*(s)$

$$V^*(s) = \max_a E[r|s,a] + \gamma \sum_{s'} \Pr(s'|s,a) V^*(s')$$

- Optimal state-action value function $7^*(s,a)$

$$Q^*(s,a) = E[r|s,a] + \gamma \sum_{s'} \Pr(s'|s,a) \max_{a'} Q^*(s',a')$$

where $V^*(s) = max_a Q^*(s,a)$

$\pi^*(s) = argmax_a Q^*(s,a)$

Monte Carlo Control

- Let G_k^a be a one-trajectory Monte Carlo target

$$G_k^a = \underbrace{r_0^{(k)}}_{a} + \underbrace{\sum_{t=1} \gamma^t r_t^{(k)}}_{\pi}$$

- Alternate between
  - **Policy evaluation**

$$Q_n^\pi(s,a) \leftarrow Q_{n-1}^\pi(s,a) + \alpha_n(G_n^a - Q_{n-1}^\pi(s,a))$$

  - **Policy improvement**

$$\pi'(s) \leftarrow argmax_a Q^\pi(s,a)$$

Q-Learning

Qlearning(s, Q^*)

Repeat

Select and execute a

Observe s' and r

Update counts: $n(s,a) \leftarrow n(s,a) + 1$

Learning rate: $\alpha \leftarrow 1/n(s,a)$

Update Q-value:

$$Q^*(s,a) \leftarrow Q^*(s,a) + \alpha(r + \gamma \max_{a'} Q^*(s',a') - Q^*(s,a))$$

$$s \leftarrow s'$$

Until convergence of Q^*

Return Q^*

|S|太大怎么办?

Q-function Approximation

- Let $s = (x_1, x_2, \ldots, x_n)^T$
- Linear

  $Q(s, a) \approx \sum_i w_{ai} x_i$
- Non-linear (e.g., neural network)

  $Q(s, a) \approx g(\boldsymbol{x}; \boldsymbol{w})$

Gradient Q-learning

- Minimize squared error between Q-value estimate and target
  - Q-value estimate: $Q_{\boldsymbol{w}}(s,a)$
  - Target: $r + \gamma \max_{a'} Q_{\bar{w}}(s',a')$
- Squared error:

$$Err(w) = \frac{1}{2}[Q_w(s,a) - r - \gamma \max_{a'} Q_{\bar{w}}(s',a')]^2$$

$\bar{w}$ fixed

- Gradient

$$\frac{\partial Err}{\partial \boldsymbol{w}} = [Q_{\boldsymbol{w}}(s,a) - r - \gamma \max_{a'} Q_{\bar{\boldsymbol{w}}}(s',a')] \frac{\partial Q_{\boldsymbol{w}}(s,a)}{\partial \boldsymbol{w}}$$

Gradient Q-learning

Initialize weights w uniformly at random in $[-1,1]$

Observe current state s

Loop

Select action a and execute it

Receive immediate reward r

Observe new state s'

Gradient: $\frac{\partial Err}{\partial \boldsymbol{w}} = [Q_{\boldsymbol{w}}(s,a) - r - \gamma \max_{a'} Q_w(s',a')] \frac{\partial Q_{\boldsymbol{w}}(s,a)}{\partial \boldsymbol{w}}$

Update weights: $\boldsymbol{w} \leftarrow \boldsymbol{w} - \alpha \frac{\partial Err}{\partial \boldsymbol{w}}$

Update state: $s \leftarrow s'$

Convergence of Linear Gradient Q-Learning

- Linear Q-Learning converges under the same conditions:

$$\sum_{n=0}^{\infty} \alpha_n = \infty \text{ and } \sum_{n=0}^{\infty} \alpha_n^2 < \infty$$

- Let $\alpha_n = 1/n$
- Let $Q_w(s,a) = \sum_i w_i x_i$

- Q-learning

$$\boldsymbol{w} \leftarrow \boldsymbol{w} - \alpha_n [Q_{\boldsymbol{w}}(s,a) - r - \gamma \max_{a'} Q_{\boldsymbol{w}}(s',a')] \frac{\partial Q_{\boldsymbol{w}}(s,a)}{\partial \boldsymbol{w}}$$

Mitigating divergence

- Two simple tricks are often used in practice:

1. Experience replay
2. Use two networks:
   - Q-network
   - Target network

- Advantages
  - Break correlations between successive updates (more stable learning)
  - Fewer interactions with environment needed to converge (greater data efficiency)

Deep Q-network

Initialize weights w and $\bar{w}$ random in $[-1,1]$

Observe current state s

Loop

Select action a and execute it

Receive immediate reward r

Observe new state s'

Add (s, a, s', r) to experience buffer

Sample mini-batch of experiences from buffer

For each experience $(\hat{s}, \hat{a}, \hat{s}', \hat{r})$ in mini-batch

Gradient: $\frac{\partial Err}{\partial \boldsymbol{w}} = [Q_{\boldsymbol{w}}(\hat{s}, \hat{a}) - \hat{r} - \gamma \max_{a'} Q_{\bar{\boldsymbol{w}}}(\hat{s}', \hat{a}')] \frac{\partial Q_{\boldsymbol{w}}(\hat{s}, \hat{a})}{\partial \boldsymbol{w}}$

Update weights: $w \leftarrow w - \alpha \frac{\partial Err}{\partial W}$

Update state: $s \leftarrow s'$

Every c steps, update target: $\bar{\boldsymbol{w}} \leftarrow \boldsymbol{w}$

Deep Q-Network for Atari